PREVUMO

PREVentive Usage MOnitor



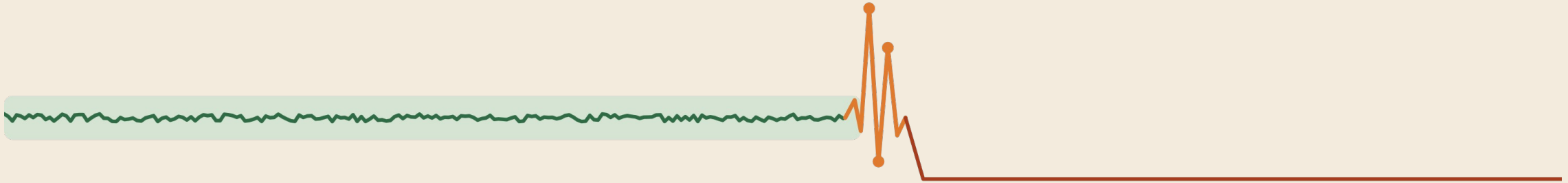
Know what happened to your machine, and when it's about to fail.

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GNG4120 · Group 10 · Oct 23, 2026

A $3,000 welder dies on a Tuesday.

Nobody saw it coming.



The owner knows who booked it. Not what happened to it.

Machines break, and nobody sees it coming

More makerspaces, labs and shared kitchens now rent out expensive electric equipment.

$754

Average emergency repair call

“Hundreds if not thousands of dollars” per breakdown, a makerspace director told us

$600 to $6,000

What each lost member is worth

When machines keep breaking, members cancel

No history

No sensor sees the warning signs

Booking tools only work “when people actually use the software”

Sources: makerspace director interview (Group 10, 2026); emergency repair call average, 86 Repairs 2024, via Envigilance.

Who we serve, and what we promise

Owners and managers of shared spaces with costly electric equipment

- Makerspaces
- University labs
- Shared kitchens

One owner can run several locations.

Value proposition

Catch failures before they happen,

know who used each machine and how much,

and shut it down safely from anywhere.

How it works

1



Box on the machine

Keypad, current sensor, relay. ESP32 with Wi-Fi.

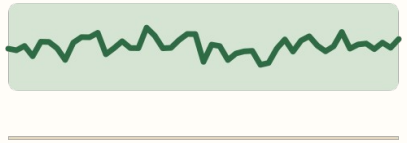
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Wi-Fi to the cloud

Every second of current, per session.

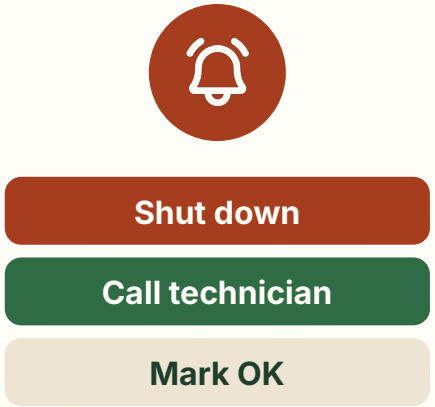
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Learns its curve

Each machine gets its own normal range.

4

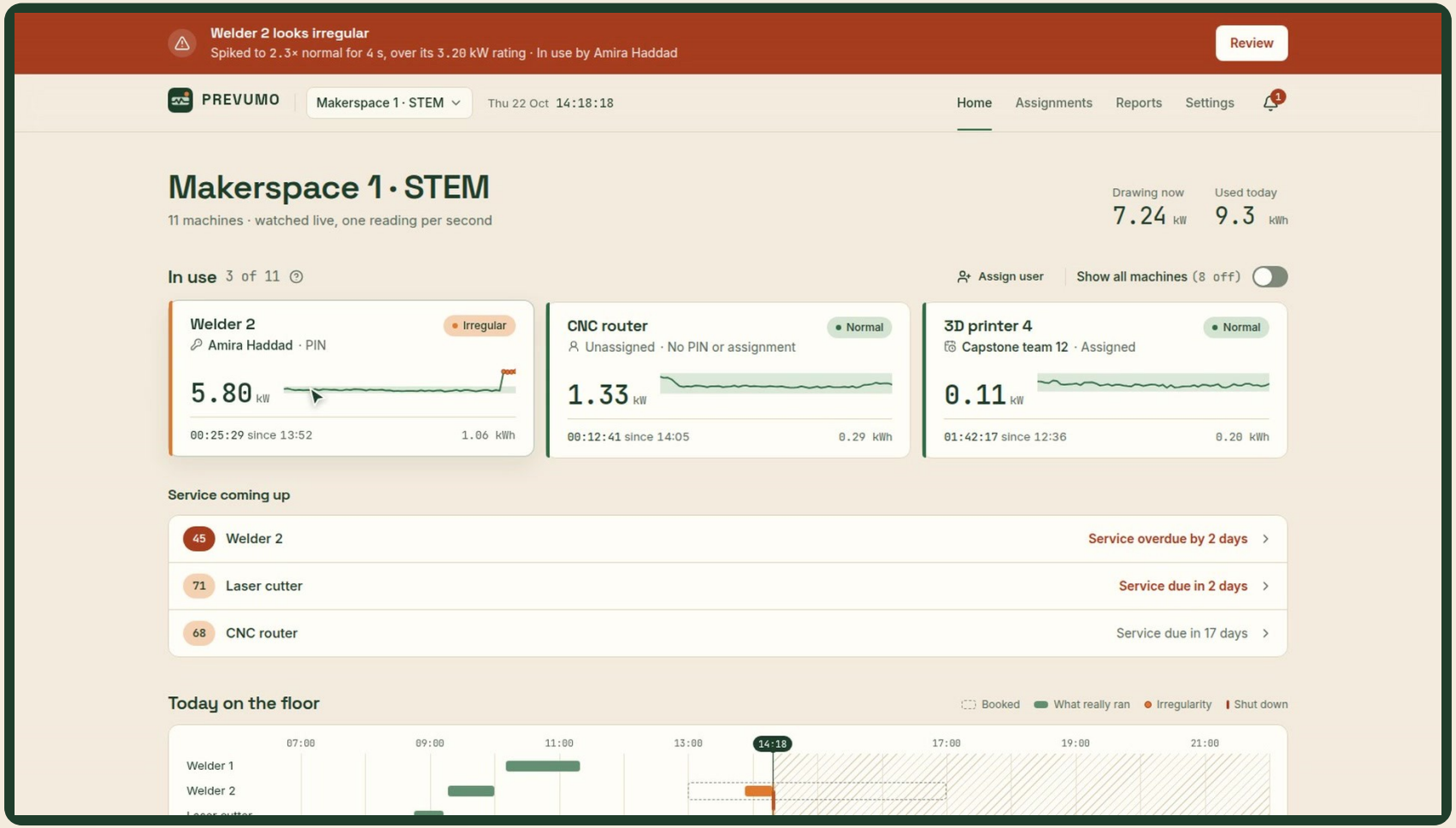


Owner decides

Alert, then one tap.

Safety: no auto restart after a shutdown, and fridges are never cut.

The owner app, live



Owner side prototype. Simulated boxes, real flow. Click to play.

Try it yourself

QR
[LIVE APP URL]

Login, alert, technician, idle shutdown, assign, daily report.

A market that runs on electric equipment

31

shared-use commercial kitchens in Ontario

poidata.io, Aug 2025

30 to 40%

saved by predictive maintenance versus fixing after breakdown

U.S. DOE, O&M Best Practices Guide

5 to 7×

more energy per sq ft in restaurants than other commercial buildings

ENERGY STAR



Expansion path

One makerspace → University networks with several labs → Shared kitchens

What each one knows

Product	Category	Price	What it knows
Skedda	Booking	US$249+ /mo	Who booked
Fabman	Machine access	€49 to €199 /mo + €10 to €14 per machine	Who switched it on
Emporia	Energy monitor	~US$200	How much energy
MachineMetrics	Factory monitoring	~US$1,400+ hardware per kit	Machine health, for factories only
PREVUMO	Shared spaces	Assumption, slide 9	**Who used it + how much + machine health + remote shutdown**

Skedda could be a partner, not just a competitor: a booking becomes a PIN on our box.

Prices from each company's public pricing, gathered in our competitor research.

From billing to predictive maintenance

Plan A (PD A)

Usage tracker and billing

Partners	Activities	Value	Relations	Segments
	Resources		Channels	
Costs		Revenue		

Plan B (now)

Safety and predictive maintenance

Partners	Activities	**Value**	Relations	Segments
	Resources		Channels	
Costs		**Revenue**		

Orange = changed

Revenue (assumptions)

$300
box per machine, one time

$25
per machine per month

+ possible referral fee on technician jobs

Key partners: electricians and technicians, component suppliers, booking platforms. Per machine pricing inspired by Fabman.

What we've tested so far

3
competitors reverse engineered

2
business model canvases

49
outreach emails, Canada, US, Europe

[N]
interviews [FILL IN]

The pivot
Billing → predictive maintenance

> **"Managing equipment is one of the biggest tasks we have."**
>
> "We are currently working on a system to track electricity usage on the welders..."
>
> Executive Director, Ontario makerspace

Assumption	Method	Result
Owners bill electricity per user	Interviews	**Invalidated**
Owners want warning before failures	Interviews	[FILL IN]
Kitchens will pay to cut idle energy	49 emails	Pending

Full validation board in the appendix.

Owners don't bill electricity per user, it's built into membership fees. So the headline moved to predictive maintenance. Billing stays as a bonus.

Skedda tells you who booked it.

PREVUMO tells you who used it, and when it's about to fail.



Thank you. Questions?

Try the app

QR
[LIVE APP URL]

Remote shutdown hardware

Up to 15 A at 120 V

A certified relay inside the box switches the machine directly.

240 V or 3-phase

The box drives a contactor, installed by a licensed electrician.

Before we sell

CSA certification of the box is required.

After any shutdown the machine stays off until someone starts it again. Fridges and freezers are never cut.

Learning, privacy and limits

About 7 days

to learn a new machine's normal current pattern. Alerts switch on after that.

Owner only

Usage data belongs to the owner's account. Members don't see each other's sessions.

Electric only

Gas equipment is invisible to electricity data.



Validation board

Customer	Problem	Core assumption	Method	Success criterion	Result
Makerspace owners	Electricity costs per member are invisible	Owners want to bill electricity per user	Interviews	[FILL IN]	**Invalidated: built into membership fees**
Makerspace and lab managers	Machines fail without warning, no history	Owners want warning before a failure	Interviews	[FILL IN]	[FILL IN]
Shared kitchen operators	Idle equipment wastes energy	Kitchens will pay to cut idle energy	49 outreach emails	[FILL IN]	Pending
[FILL IN]					

Q&A prep

Why not a smart plug?

No PIN, no per member history, no learned normal curve, and most can't carry a welder.

Heavy 240 V equipment?

The box drives a contactor installed by a licensed electrician.

Liability for a shutdown?

The owner decides, every shutdown is logged, nothing restarts on its own.

How does it know normal?

About 7 days of the machine's own current, then it flags sustained changes.

Can a non-profit pay $25?

One avoided emergency call ($754) covers about 2.5 machine-years. Price still to validate.

Versus Fabman?

Fabman knows who switched it on. We add machine health, idle and technicians.

What did interviews change?

No per user billing, so predictive maintenance became the headline.

Full answers in the speaker notes.

Sources

- Makerspace director and Executive Director, Ontario makerspace: Group 10 interviews, 2026. Quotes anonymized.
- Emergency repair call average, $754: 86 Repairs, 2024, via Envigilance.
- 31 shared-use commercial kitchens in Ontario: poidata.io, August 2025.
- Predictive maintenance saves about 30 to 40% versus reactive maintenance: U.S. Department of Energy, Operations & Maintenance Best Practices Guide.
- Restaurants use 5 to 7 times more energy per square foot than other commercial buildings: ENERGY STAR.
- Competitor prices: Skedda, Fabman, Emporia and MachineMetrics public pricing, from our competitor research.
- Prices for PREVUMO ($300 box, $25 per machine per month) are assumptions to validate.